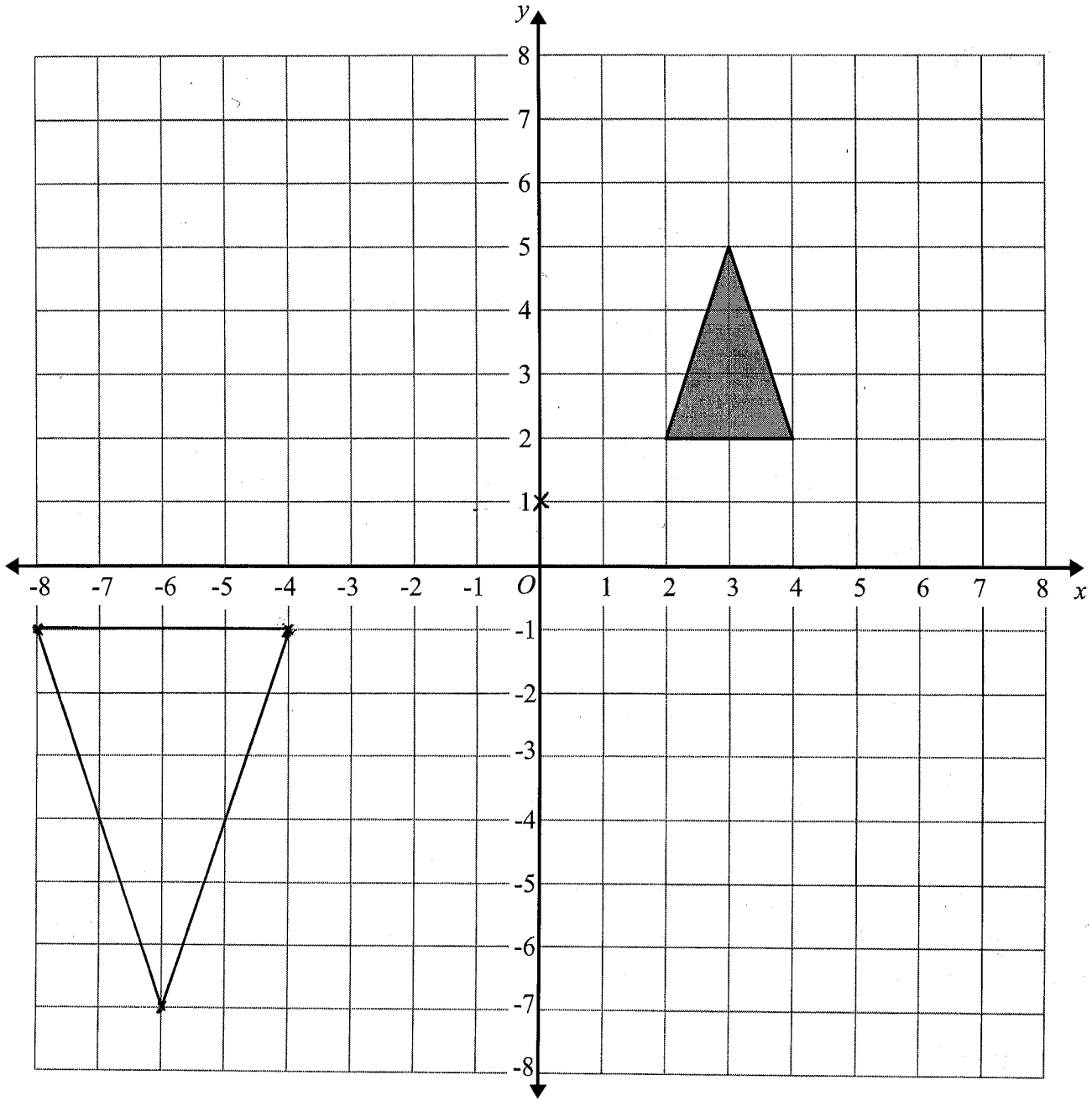
1



On the grid, enlarge the triangle by scale factor -2. centre (0,1)

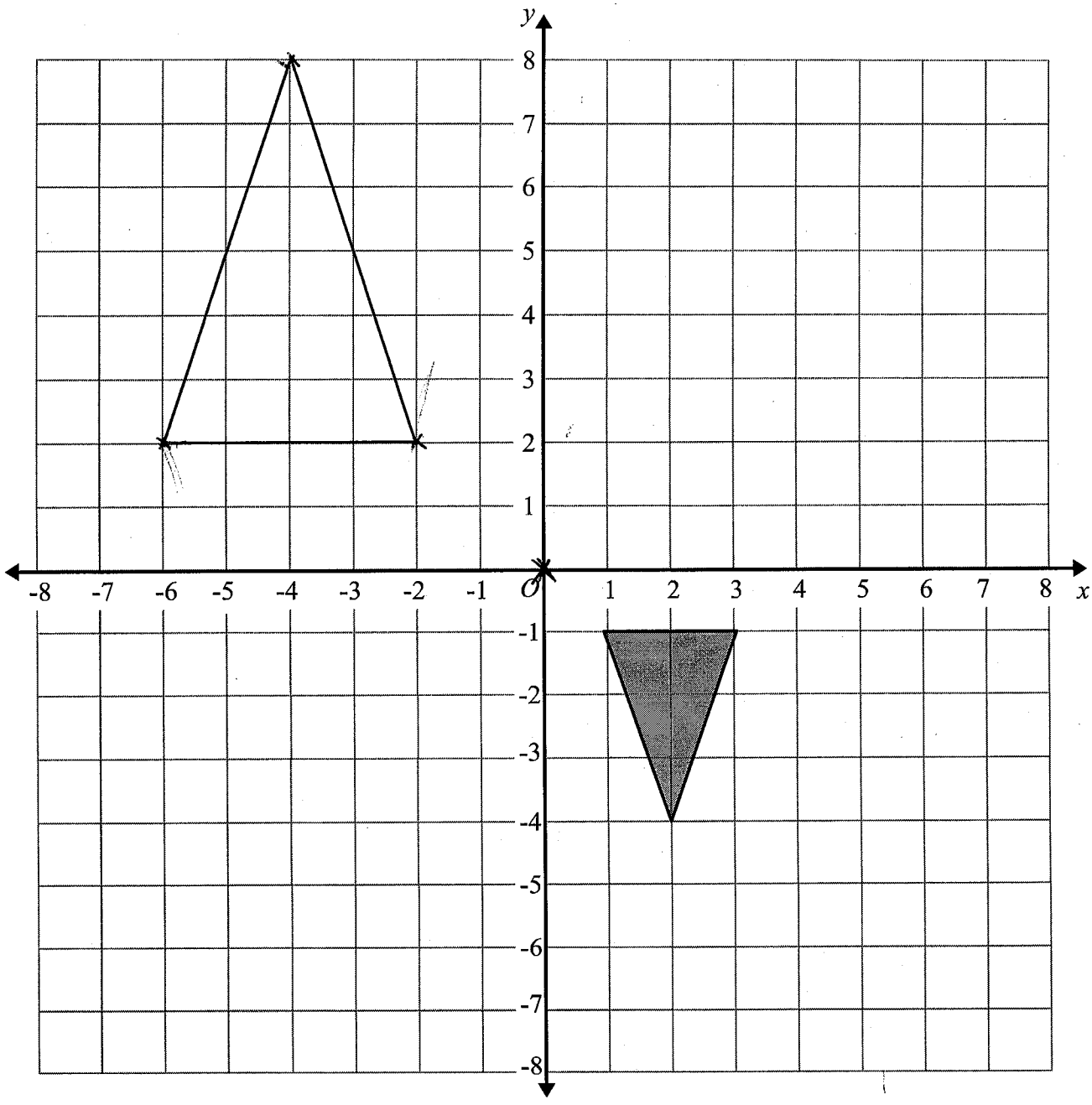
(Total for Question 1 is 2 marks)

$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \times -2 = \begin{pmatrix} -4 \\ -2 \end{pmatrix}$ From the centre

$\begin{pmatrix} 4 \\ 1 \end{pmatrix} \times -2 = \begin{pmatrix} -8 \\ -2 \end{pmatrix}$

$\begin{pmatrix} 3 \\ 4 \end{pmatrix} \times -2 = \begin{pmatrix} -6 \\ -8 \end{pmatrix}$

2



On the grid, enlarge the triangle by scale factor -2. centre *O*

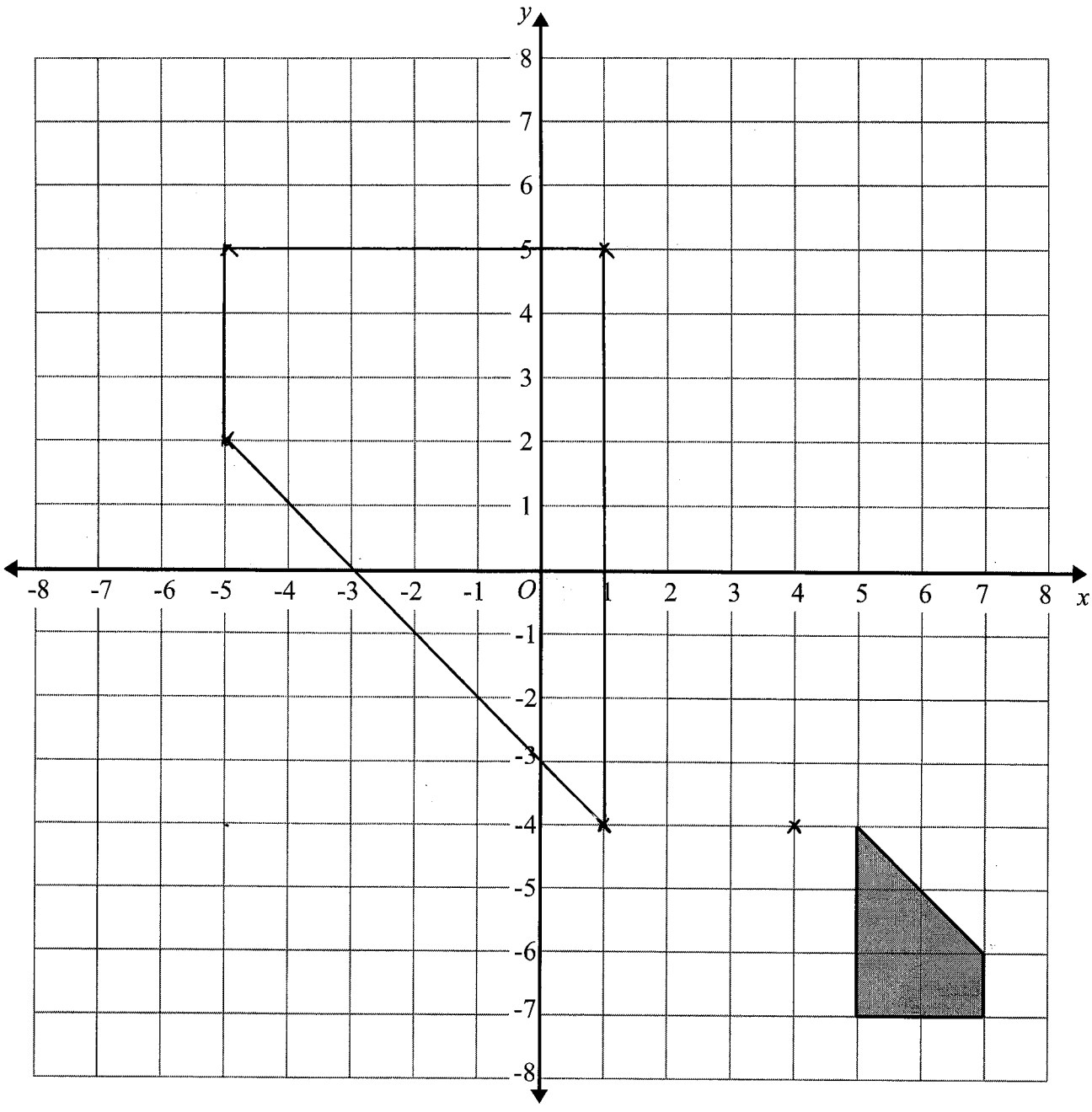
(Total for Question 2 is 2 marks)

$\begin{pmatrix} 1 \\ -1 \end{pmatrix} \times -2 = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$

$\begin{pmatrix} 3 \\ -1 \end{pmatrix} \times -2 = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$

$\begin{pmatrix} 2 \\ -4 \end{pmatrix} \times -2 = \begin{pmatrix} -4 \\ 8 \end{pmatrix}$

3



On the grid, enlarge the shape by scale factor -3. centre (4,-4)

(Total for Question 3 is 2 marks)

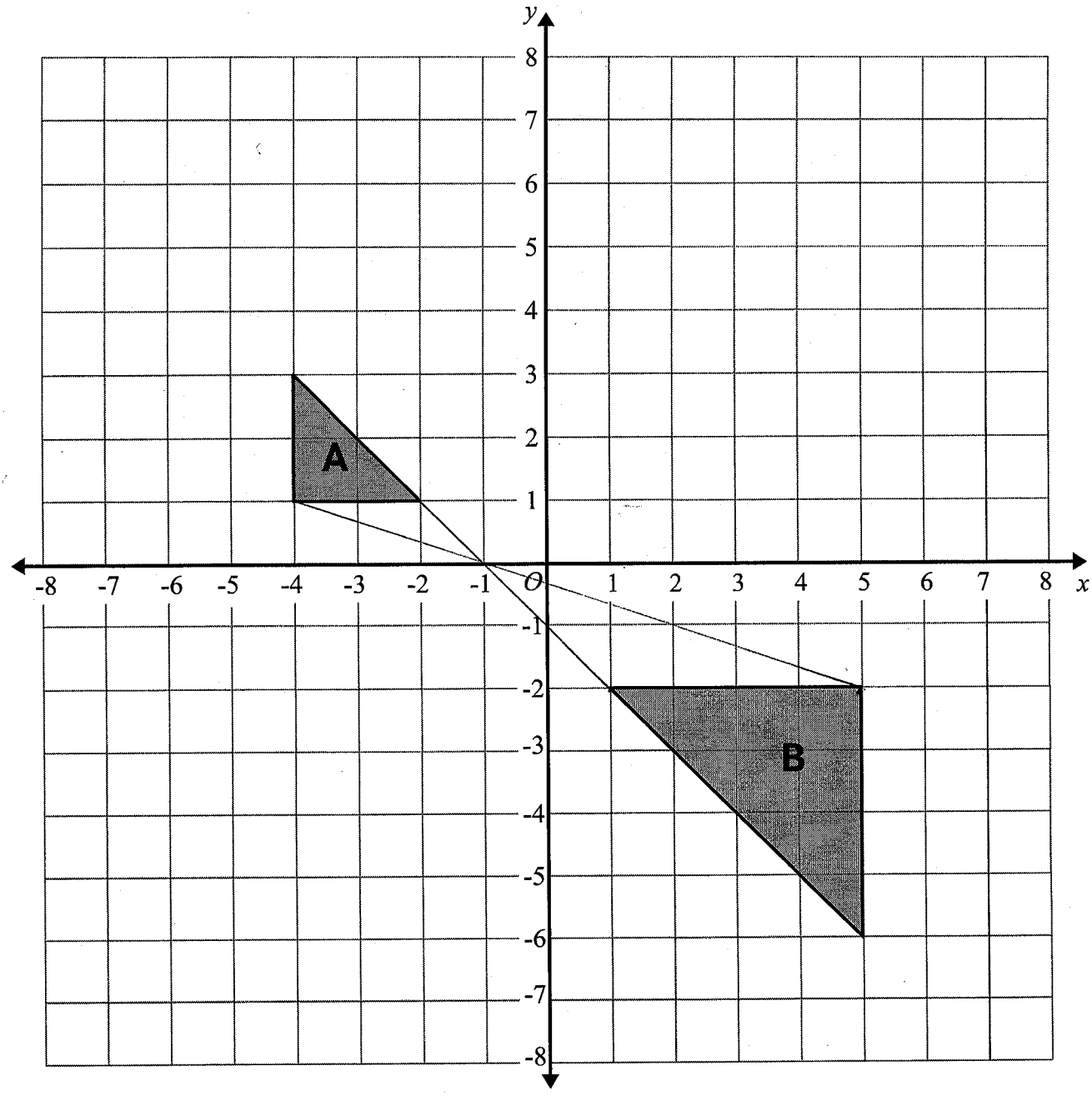
$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \times -3 = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 1 \\ -3 \end{pmatrix} \times -3 = \begin{pmatrix} -3 \\ 9 \end{pmatrix}$

$\begin{pmatrix} 3 \\ -2 \end{pmatrix} \times -3 = \begin{pmatrix} -9 \\ 6 \end{pmatrix}$

$\begin{pmatrix} 3 \\ -3 \end{pmatrix} \times -3 = \begin{pmatrix} -9 \\ 9 \end{pmatrix}$

4



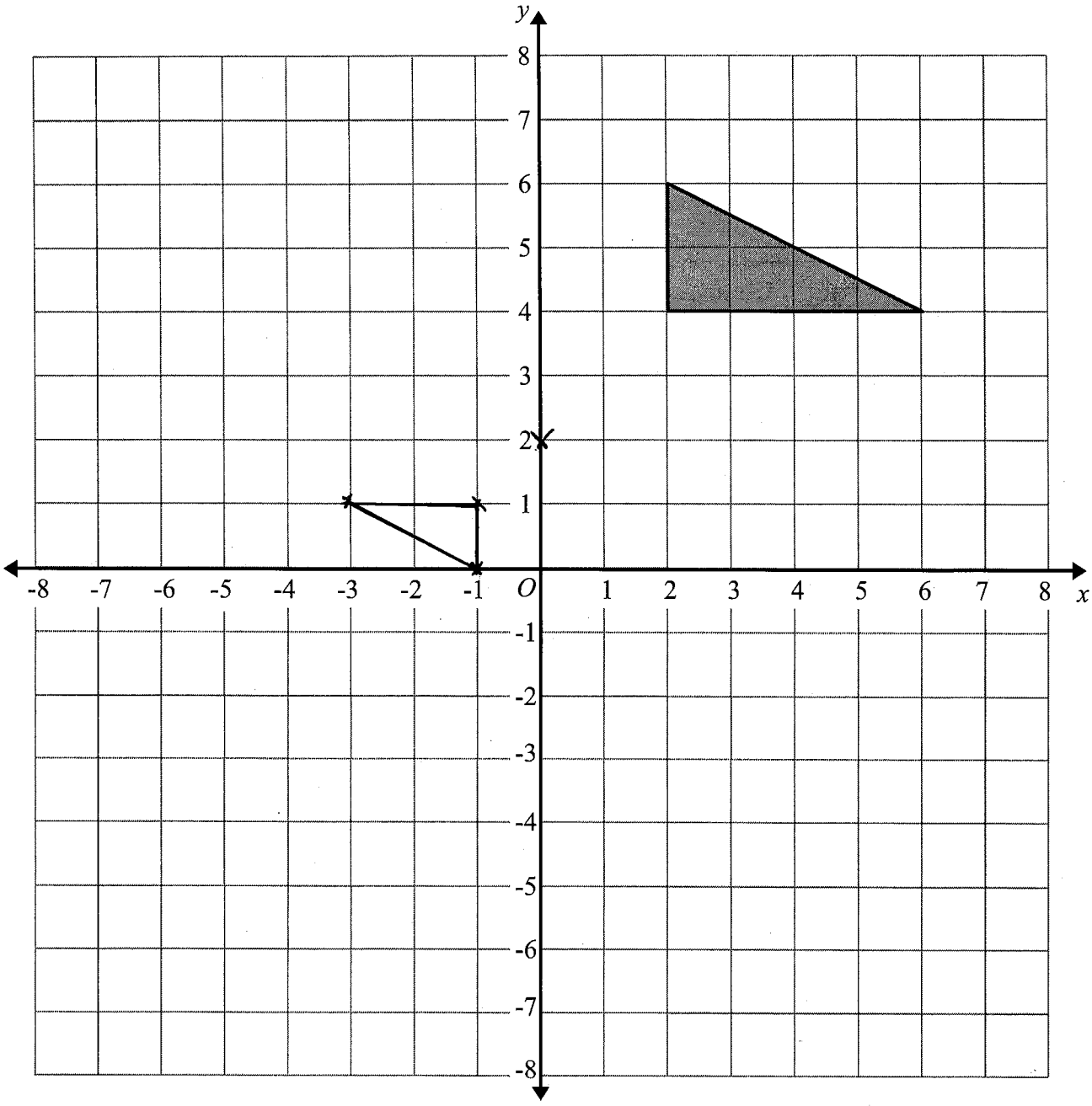
Describe fully the single transformation which maps triangle A onto triangle B.

Enlargement, Scale Factor −2

Centre (−1, 0)

(Total for Question 4 is 2 marks)

5



On the grid, enlarge the triangle by scale factor -0.5. centre (0,2)

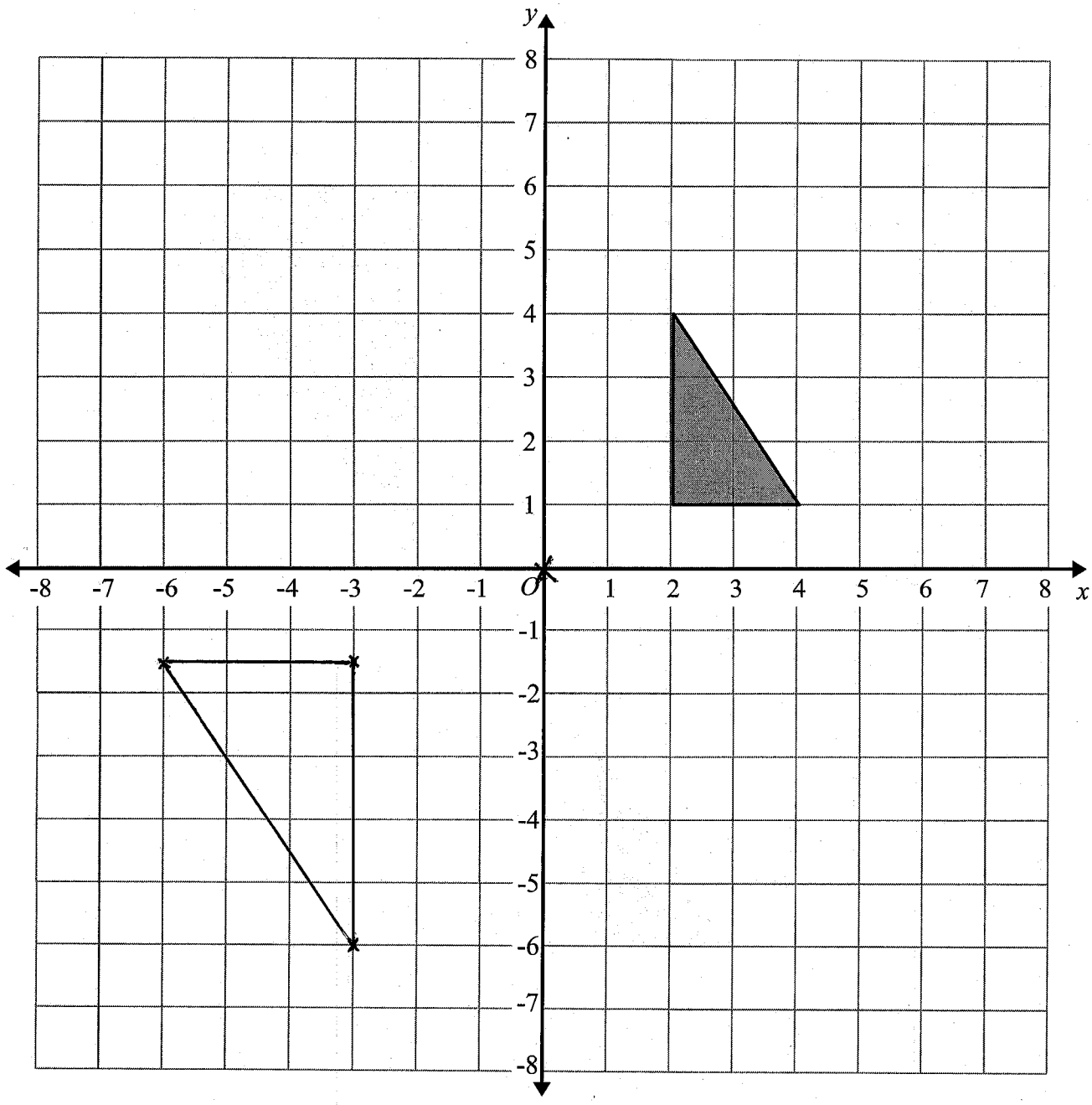
(Total for Question 5 is 2 marks)

$\begin{pmatrix} 2 \\ 2 \end{pmatrix} \times -0.5 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 4 \end{pmatrix} \times -0.5 = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$

$\begin{pmatrix} 6 \\ 2 \end{pmatrix} \times -0.5 = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$

6



On the grid, enlarge the triangle by scale factor -1.5. centre *O*.

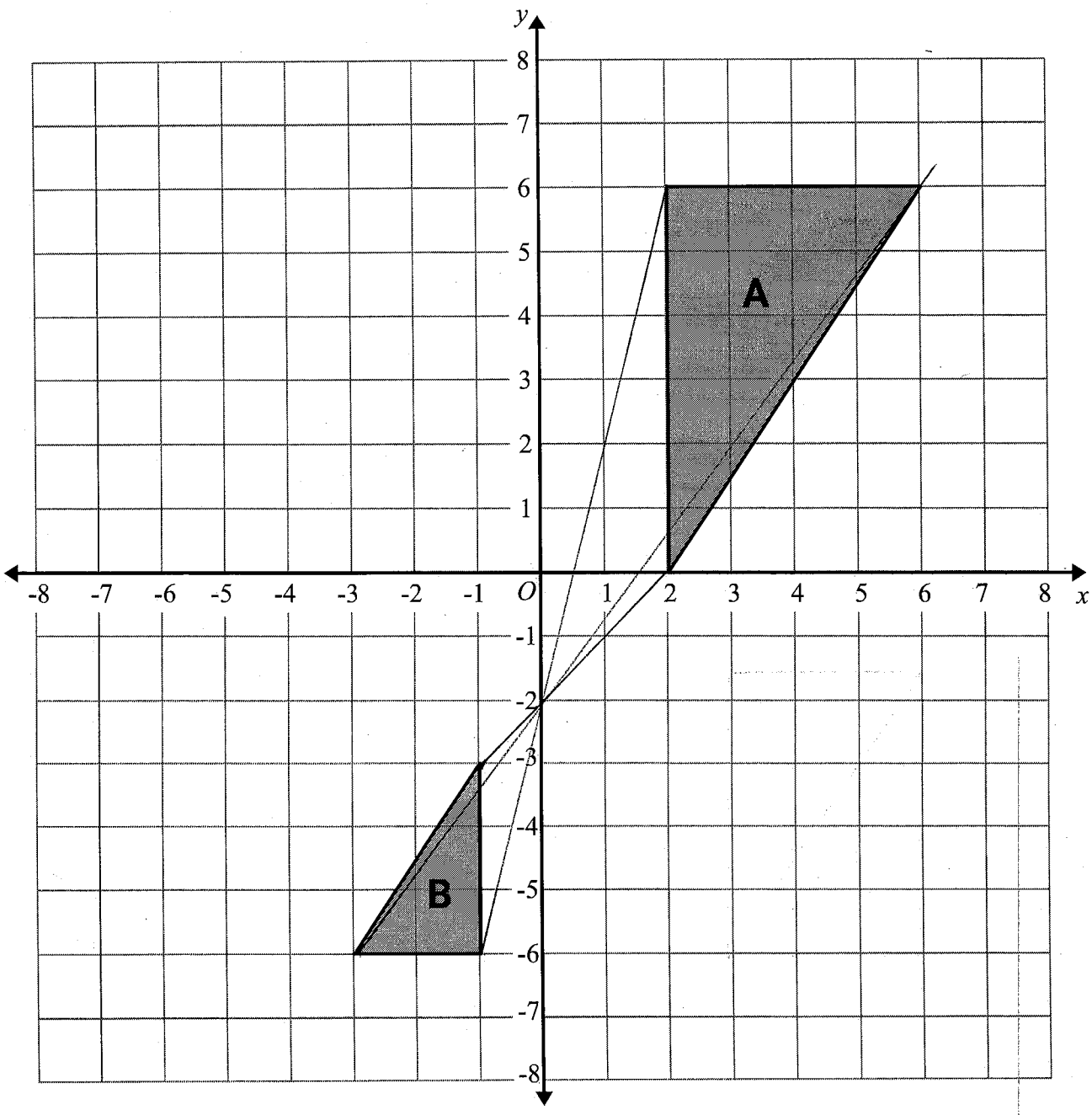
(Total for Question 6 is 2 marks)

$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \times -1.5 = \begin{pmatrix} -3 \\ -1.5 \end{pmatrix}$

$\begin{pmatrix} 4 \\ 1 \end{pmatrix} \times -1.5 = \begin{pmatrix} -6 \\ -1.5 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 4 \end{pmatrix} \times -1.5 = \begin{pmatrix} -3 \\ -6 \end{pmatrix}$

7

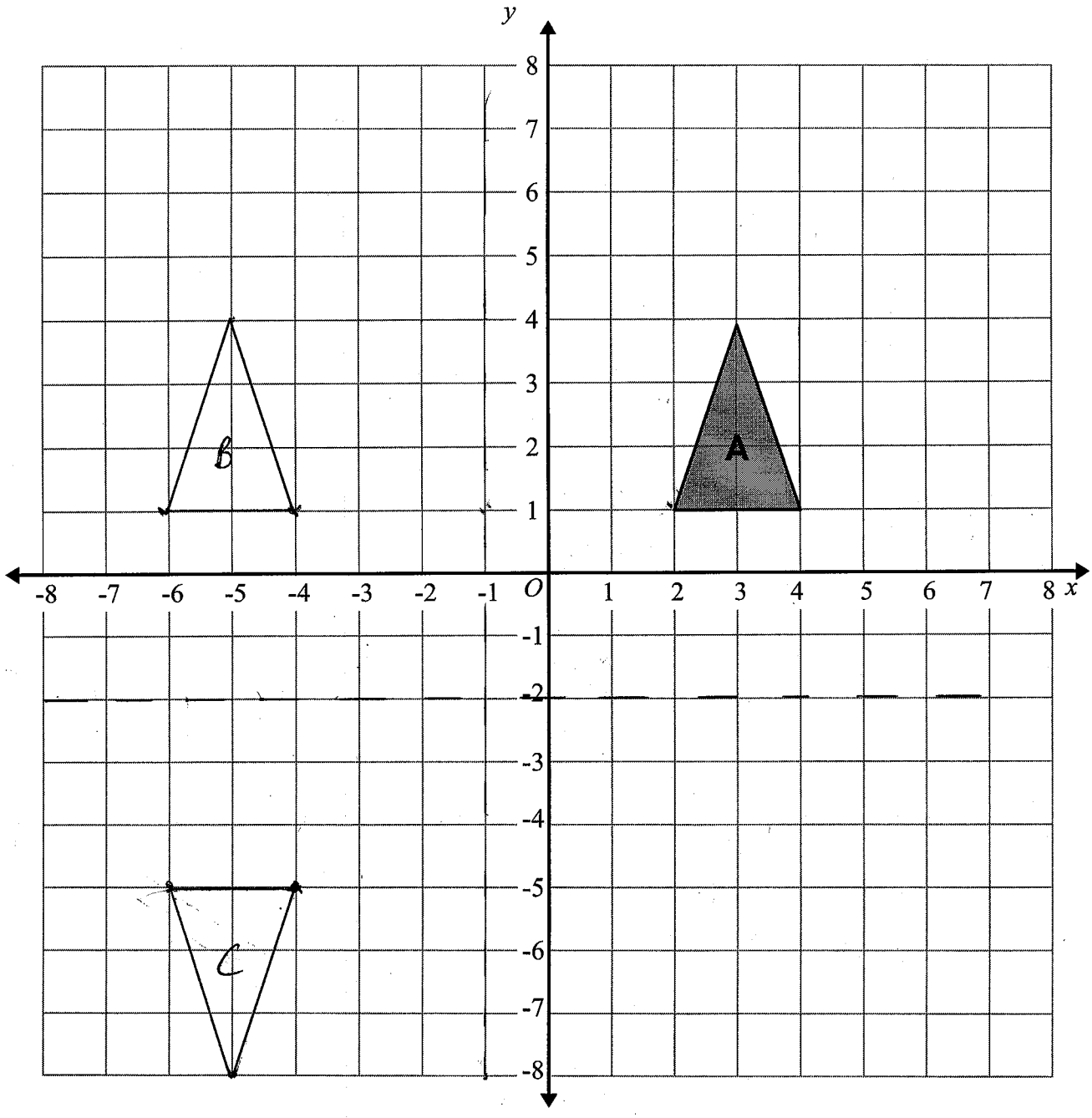


Describe fully the single transformation which maps triangle A onto triangle B.

Enlargement, Scale Factor −0.5
Centre (0, −2)

(Total for Question 7 is 2 marks)

8



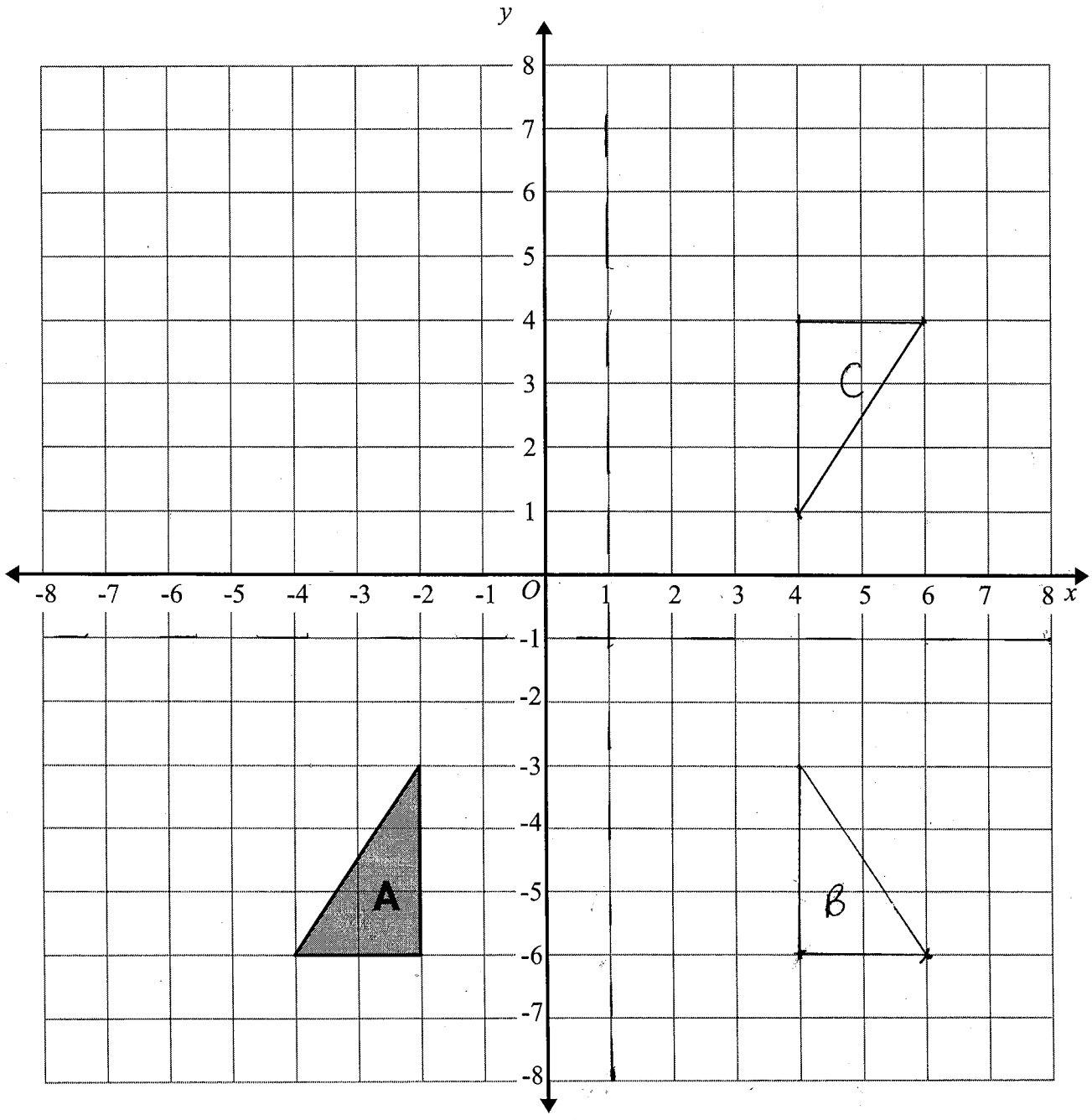
Shape **A** is reflected in the line $x = -1$ to give shape **B**.
Shape **B** is reflected in the line $y = -2$ to give shape **C**.

Describe the **single** transformation that will map shape **A** to shape **C**.

Rotation, 180°, centre (−1, −2)

(Total for Question 8 is 2 marks)

9



Shape **A** is reflected in the line $x = 1$ to give shape **B**.
Shape **B** is reflected in the line $y = -1$ to give shape **C**.

Describe the **single** transformation that will map shape **A** to shape **C**.

Rotation, 180°, centre (1, -1)

(Total for Question 9 is 2 marks)